MIL-P-52469B(ME)
19 December 1984
SUPERSEDING
MIL-P-52469A(ME)
7 December 1977

MILITARY SPECIFICATION

PUMP, CENTRIFUGAL, WATER, DIESEL-ENGINE-DRIVEN, 2-1/2-INCH, SKID-MOUNTED, 200 GPM AT 300 FEET TOTAL HEAD AUXILIARY PRIMING

This specification is approved for use by the USA Belvoir Research and Development Center, Department of the Army, and is available for use by all Departments and Agencies of the Department of Defense.

1. SCOPE

1.1 Scope. This specification covers a skid-mounted, diesel-engine-driven, auxiliary priming, centrifugal pumping assembly for water pumping service.

2. APPLICABLE DOCUMENTS

2.1 Government documents.

2.1.1 Specifications and standards. Unless otherwise specified, the following specifications and standards of the issue listed in that issue of the Department of Defense Index of Specifications and Standards (DoDISS) specified in the solicitation, form a part of this specification to the extent specified herein.

MILITARY

MIL-P-514	- Plates, Identification, Instruction and Marking, Blank.
MIL-T-704	- Treatment and Painting of Materiel.
MIL-L-2104	- Lubricating Oil, Internal Combustion Engine, Tactical Service.
MIL-F-3541	- Fittings, Lubrication.
MIL-G-3859	- Grease Guns, Hand-Operated, Lever, Push and Screw Type.
MIL-P-10603	- Pumps and Pumping Units, Centrifugal, Power Driven, for Water, Packaging of.

Beneficial comments (recommendations, additions, deletions) and any pertinent data which may be of use in improving this document should be addressed to: USA Belvoir Research and Development Center, ATTN: STRBE-DS, Fort Belvoir, VA 22060-5606 by using the self-addressed Standardization Document Improvement Proposal (DD Form 1426) appearing at the end of this document or by letter.

FSC 4320

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MIL-G-10924 - Grease, Automotive and Artillery.
MIL-G-20241 - Gasket Material, Wool Felt, Impregnated, Adhesive, Pressure-Sensitive.
MIL-P-23377 - Primer Coatings: Epoxy-Polyamide, Chemical and Solvent Resistant.
MIL-G-23827 - Grease, Aircraft and Instrument, Gear and Actuator Screw.
MIL-L-46167 - Lubricating Oil, Internal Combustion Engine, Arctic.
MIL-S-81733 - Sealing and Coating Compound, Corrosion Inhibitive.

STANDARDS

FEDERAL

FED-STD-H28 - Screw-Thread Standards for Federal Services.

MILITARY

MIL-STD-105 - Sampling Procedures and Tables for Inspection By Attributes.
MIL-STD-130 - Identification Marking of US Military Property.
MIL-STD-209 - Slinging and Tiedown Provisions for Lifting and Tying down Military Equipment.
MIL-STD-461 - Electromagnetic Interference Characteristics Requirements for Equipment.
MIL-STD-781 - Reliability Tests: Exponential Distribution.
MIL-STD-889 - Dissimilar Metals.
MIL-STD-810 - Environmental Test Methods.
MIL-STD-1400 - Engine, Gasoline and Diesel, Method of Test.
MIL-STD-1410 - Methods for Selection of Industrial Engines for End Item Application.
MIL-STD-1472 - Human Engineering Design Criteria for Military Systems, Equipment and Facilities.
MIL-STD-1474 - Noise Limits for Army Materiel.
MS35000 - Battery, Storage, Lead-Acid, Waterproof.
MS35644 - Filler Neck Assembly, Fuel Tank-Military Vehicles.
MS35645 - Cap, Fuel Tank: Tactical Vehicle, Fording.
MS53063 - Indicator, Air Cleaner, Intake, Restriction, Mechanical Type.

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2.1.2 Other Government documents, drawings, and publications. The following other Government documents, drawings, and publications form a part of this specification to the extent specified herein.

ENVIRONMENTAL PROTECTION AGENCY (EPA)

Title 40, Code of Federal Regulations, Part 85 - Control of Air Pollution from New Motor Vehicles and New Motor Vehicle Engines.

(Application for copies should be addressed to the Superintendent of Documents, US Government Printing Office, Washington, DC 20402.)

(Copies of specifications, standards, and drawings required by contractors in connection with specific acquisition functions should be obtained from the contracting activity or as directed by the contracting officer.)

2.2 Other publications. The following document(s) form a part of this specification to the extent specified herein. The issues of the documents which are indicated as DoD adopted shall be the issue listed in the current DoDISS and the supplement thereto, if applicable.

AMERICAN NATIONAL STANDARD INSTITUTE INC. (ANSI)

B16.1 - Cast Iron Pipe Flanges and Flanged Fittings.

(Application for copies should be addressed to the American National Standards Institute, 1430 Broadway, New York, NY 10018.)

AMERICAN SOCIETY FOR TESTING AND MATERIALS (ASTM)

A 48 - Gray Iron Castings.
A 276 - Stainless and Heat-Resisting Steel Bars and Shapes.
A 511 - Seamless Stainless Steel Mechanical Tubing.
B 148 - Aluminum-Bronze Sand Castings.
B 308 - Aluminum-Alloy Standard Structural Shapes, Rolled or Extruded.
B 584 - Copper-Alloy Sand Casting for General Applications.
D 3951 - Standard Practice for Commercial Packaging.

(Application for copies should be addressed to the American Society for Testing and Materials, 1916 Race Street, Philadelphia, PA 19103.)

HYDRAULIC INSTITUTE

Rating Standards and Test Code Standards.

(Application for copies should be addressed to the Hydraulic Institute, 2130 Keith Building, Cleveland, OH 44115.)

SOCIETY OF AUTOMOTIVE ENGINEERS (SAE)

SAE Handbook

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(Application for copies should be addressed to the Society of Automotive Engineers, 400 Commonwealth Drive, Warrendale, PA 15096.)

STATE OF CALIFORNIA AIR RESOURCES BOARD

Health and Safety Code.

(Application for copies should be addressed to the California Air Resources Board, 1709 Eleventh Street, Sacramento, CA 95814.)

(Industry association specifications and standards are generally available for reference from libraries. They are also distributed among technical groups and using Federal agencies.)

2.3 Order of precedence. In the event of a conflict between the text of this specification and the references cited herein, the text of this specification shall take precedence.

3. REQUIREMENTS

3.1 Description. The pump assembly shall consist of skid-mounted centrifugal pump, diesel engine, and all necessary accessories for pumping water.

3.2 First article. When specified (see 6.2), a sample shall be subjected to first article inspection (see 4.3 and 6.3). Any changes or deviations of pumping assemblies from the approved first article during production will be subject to the approval of the contracting officer. Approval of the first article will not relieve the contractor of his obligation to furnish pumping assemblies conforming to this specification.

3.3 Material. Material shall be as specified herein. Materials not specified shall be selected by the contractor and shall be subject to all provisions of this specification.

3.3.1 Material deterioration and control. The pump assembly shall be fabricated from compatible materials, inherently corrosion resistant or treated to be resistant against the various forms of corrosion and deterioration that may be encountered in any of the applicable storage and operating environments to which the item may be exposed.

3.3.1.1 Dissimilar metals. Dissimilar metals as defined in MIL-STD-889, shall be electrically insulated from one another to minimize or prevent galvanic corrosion. Insulation shall be provided by an insulating barrier such as a coat of epoxy primer conforming to MIL-P-23377, or corrosion inhibiting sealant conforming to MIL-S-81733 or chromate tape conforming to MIL-G-20241. Protection against any galvanic corrosion could also be obtained by exclusion of the electrolyte if feasible.

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3.3.1.2 Identification of materials and finishes. The contractor shall identify the specific material, material finish or treatment for use with components and sub-components, and shall make information available, upon request to the contracting officer or designated representative.

3.3.1.3 Recovered materials. For the purpose of this requirement, recovered materials are those materials which have been collected from solid waste and reprocessed to become a source of raw materials, as distinguished from virgin raw materials. The components, pieces and parts incorporated in the pumping assembly may be newly fabricated from recovered materials to the maximum extent practicable, provided the pumping assembly produced meets all other requirements of this specification. Used, rebuilt or remanufactured components, pieces and parts shall not be incorporated in the pumping assembly .

3.4 Threads. Screw threads shall conform to FED-STD-H28.

3.5 Environmental conditions.

3.5.1 Storage temperatures. The pump assembly shall be capable of withstanding storage without damage in a temperature of -30° F to +160° F. The pump assembly shall be tested for performance in accordance with 4.6.2.3 and 4.6.2.4.

3.6 Safety. All shafts or rotating or moving parts of the pump assembly shall be guarded when such parts are exposed to contact by personnel or otherwise create a hazard. All hot surfaces of the equipment, including exhaust pipes exposed to contact by personnel or which create a fire hazard, shall be fully guarded or insulated. Fuel tanks shall be located in a manner which will not allow spills or overflows to run onto the engine, exhaust, or electrical equipment. Exhaust or discharges from the equipment shall be directed so that they do not endanger personnel or obstruct the view of the operator. Electrical equipment shall be effectively guarded and grounded to protect all persons and objects from electrical shock hazard. Battery compartments shall be well ventilated and battery terminals shall be guarded to prevent accidental shorting.

3.7 Human factors engineering. The pumps shall conform to accepted human factors engineering design criteria as described in MIL-STD-1472. Special design emphasis shall be given, but not limited to, section 4 (General Requirements), 5.1 (Control/Display Integration), 5.2 (Visual Displays), 5.4 (Controls), 5.5 (Labeling), 5.6 (Anthropometry), 5.9 (Design for Maintainability), and 5.13 (Hazards and Safety) of MIL-STD-1472 as applicable. The pumps shall be operable and maintainable by 5th through 95th percentile soldiers appropriately clothed for the environments of interest.

3.7.1 Noise limits. The noise produced by the pumps shall not exceed MIL-STD-1474, category D [85 dB(A)] at the operator's position or other occasionally occupied positions when tested in accordance with 4.6.2.6.1. If technical infeasability is established via MIL-STD-1474, requirement for maximum noise limits, or if the means of reduction is too costly, the written permission of the procuring activity shall be requested to exceed the 85 dB(A) limit. If the

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procedures of MIL-STD-1474 requirements for maximum noise limits have been pursued and documented to the satisfaction of the procuring activity, then the precautions of MIL-STD-1474 requirement for the posting noise hazard areas and manuals shall be provided. Hazard signs if required shall conform with MIL-STD-1474 and be readable at the distance indicated on the sign.

3.8 Reliability. The specified mean-time-between-failure (MTBF) shall be 200 hours when the pump assembly is tested as specified in 4.6.2.7.

3.9 Maintainability.

3.9.1 Maintenance support. Assemblages or support elements such as technical manuals, repair parts, special tools, or lubrication charts, as specified (see 6.4 and 6.5), shall be adequate to perform the intended function of assisting or conducting maintenance operations when tested as specified in 4.6.2.8.1.

3.9.2 Maintenance ratio. The pump assembly shall have a maintenance ratio of not more than 0.07. Maintenance ratio is defined as the ratio of the total active maintenance man-hours required (scheduled and unscheduled) to the total operating time. Man-hours for repair of replaced components and scheduled before-and-after operation checks are excluded. A maintenance schedule shall be furnished prior to the start of any testing.

3.10 Performance. When operated at a speed not exceeding the maximum continuous speed of the engine as recommended by the engine manufacturer, with a lift of 10 feet through 20 feet of 2-1/2-inch suction hose, the pump assembly shall prime and discharge water as follows:

a. Deliver a minimum of 200 gallons per minute (gpm) at 300-foot total head when operated under a 10-foot, static suction lift at not more than the manufacturer's recommended governed speed of the engine.
b. Deliver a minimum of 170 gpm when operated under a 20-foot, static suction lift at not more than the manufacturer's recommended governed speed of the engine.

3.10.1 Performance identification plate. A performance identification plate not less than 8 inches by 10-1/2 inches shall be permanently attached to the inside cover of the instrument panel. The chart shall show the complete performance characteristics of the pump unit when operating at the maximum continuous operating speed. All data shall be derived from the actual performance of the first article. The plate shall conform to MIL-P-514, type III, composition C of type I, grade A, class 1 material.

3.11 Pump. The pump shall be the straight centrifugal, single-suction, single-stage, vertical split-case, auxiliary priming type. The maximum net positive suction head (NPSH) requirement shall not exceed 9 feet of water at 225 gpm at 350 feet total head at not more than the manufacturer's recommended speed of the engine. The pump shall have an aluminum-bronze impeller and wearing rings. The impeller shaft shall be corrosion-resisting steel. All pump shaft glands or mechanical seals shall be adequate for service under operating conditions specified herein. When necessary, a gear box may be used to establish the correct speed ratio between the engine and the pump.

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3.11.1 Hydrostatic pressure. The pump with check valve assembly installed shall not leak or show permanent deformation when subjected to an internal hydrostatic pressure of not less than 260 pounds per square inch (psi).

3.11.2 Pump case. The pump case shall be a vertical split-iron casting conforming to ASTM A 48, class 40. The pump casting shall be provided with a drain port and plug. The drain port shall be internally threaded NPT and shall be of such size and location so as to completely drain the pump case. When the plug is located in the volute passage, the plug shall not protrude beyond the inner surface of the pump case.

3.11.3 Pump impeller. The pump impeller shall be cast aluminum-bronze conforming to ASTM B 148, copper alloy UNS No. C95500, and shall be single-suction, fully enclosed type. The impeller shall be designed to mount on the impeller shaft.

3.11.3.1 Impeller balancing. The impeller shall be dynamically balanced. The "out-of-balance" condition shall be not more than 0.25 ounce-inch (oz-in).

3.11.4 Impeller shaft sleeve. The pump seal shall be mounted on a shaft sleeve. The shaft sleeve shall be corrosion-resisting steel conforming to ASTM A 511, grade MT304, cold-worked, seamless steel mechanical tubing with a yield strength of not less than 85,000 psi. The sleeve may also be made from corrosion-resisting steel conforming to ASTM A 276, UNS S30400, condition B.

3.11.5 Wearing rings. The replaceable wearing rings shall be copper alloy conforming to ASTM B 584, copper alloy UNS No. C83600.

3.11.6 Pump suction port. The suction port of the pump shall be provided with an integrally cast 2-1/2-inch NPT companion flange conforming to ANSI B16.1.

3.11.7 Pump discharge port. The discharge port of the pump shall be provided with a cast iron 2-inch nominal pipe flange conforming to ANSI B16.1. The flange of the pump shall mate with the companion flange of the check valve assembly.

3.11.8 Check valve assembly. A check valve assembly shall be provided on the discharge side of the pump. The check valve shall prevent the reverse flow of liquid from the discharge line back through the pump case. The valve assembly shall be a 2-inch nominal pipe flange conforming to ANSI B16.1 except that the flange body shall be not less than 1/2-inch thick on one end and 2-1/2-inch NPT internal threads on the other end. The flanged end of the valve assembly shall mount to the mating flange of the pump discharge port.

3.11.9 Priming. When operating at conditions adjusted to be equivalent to standard sea level conditions of 29.92 inches of mercury, using water with a specific weight of 62.3 pounds per cubic foot at 68° F, and using 15 feet or 25 feet of suction line (as applicable), containing one elbow, the pump shall meet the following priming requirements:

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a. Prime in not more than 30 seconds under a 10-foot static suction lift.
b. Prime in not more than 1 minute under a 20-foot static suction lift.

3.12 Diesel engine. For the purpose of selecting the engine, the pump unit shall be classified as a class II end item in accordance with MIL-STD-1410. The engine, including all systems, components, and accessories, shall conform to MIL-STD-1410, unless otherwise specified herein. The maximum governed speed for each engine, as applied, shall not exceed the engine manufacturer's recommended speed for this application. All engine accessories shall be furnished by or approved by the engine manufacturer for this application. The engine shall be capable of operating on fuels and lubricating oils as specified in MIL-STD-1410. The engine shall comply with the Environmental Protection Agency and the California Air Resources Board regulations as applicable governing control of air pollution for new motor vehicles and new engines, in effect for this type of equipment on date of manufacture. The engine shall meet all operational requirements and conditions specified herein without exceeding the limits specified in table I of MIL-STD-1410, or, the engine manufacturer's recommended operating limits. The average oil consumption of each engine shall not exceed .0035 pounds per brake-horsepower-hour (bhp/hr) when tested as specified herein.

3.12.1 Engine control systems. The engine shall be furnished with control systems in accordance with MIL-STD-1410. The control systems shall include a high-temperature control, low-oil-pressure control, and overspeed control. All controls shall be of the engine-shut-off type.

3.12.2 Engine operation requirements. The engine operational requirements and maximum engine operating limits shall be in accordance with MIL-STD-1410 under any of the requirements and conditions specified herein. The engine, as applied to the pump unit, shall meet the applicable State of California exhaust emissions standards.

3.12.3 Fuel system. A fuel tank shall be furnished. The capacity of the tank shall be sufficient for not less than 4 hours of continuous operation at the rated load. The fuel tank shall be equipped with a cap conforming to MS35645 and a strainer and neck conforming to MS35644. The tank shall be provided with a hand-operated fuel shutoff valve attached directly to the tank and a mechanically actuated fuel gage mounted on the tank. The fuel tank outlet shall be not less than 1/2-inch above the bottom of the tank. The fuel line shall be 5/16-inch OD and shall be provided with a 3-way selector valve and a 1/2-inch 20 SAE flared fitting for connection with an auxiliary fuel supply. A corrosion-resisting circular metal face-plate shall be mounted on the fuel selector valve and shall be marked "OFF", "UNIT TANK" and "AUXILIARY". The fuel tank shall not leak when tested as specified in 4.6.2.14.

3.12.4 Instrumentation. An instrument panel shall be furnished and shall be equipped with the following instruments:

a. Gage, pressure, engine lubricating oil.
b. Tachometer and hourmeter conforming to SAE J678d.
c. Ammeter, battery charging.
d. Engine coolant temperature gage (if liquid cooled).
e. Air cleaner restriction indicator in accordance with MS53063.

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3.12.5 Manual engine controls. Manual engine controls shall include a fuel priming system, cold starting control, ether starting aid if required, switch, and throttle. The controls shall be mounted on the same panel as the instruments and shall be clearly and permanently marked as to their function.

3.12.6 Cranking system. An electrical cranking system based on the use of a 12-volt electrical circuit energized by storage batteries shall be furnished.

3.12.7 Charging system. A rectified alternating-current-type battery-charging system shall be provided. Batteries shall conform to MS35000, type 2HN. The battery-charging alternator shall have a rating of not less than 35 amperes. Two batteries shall be furnished, connected in parallel.

3.13 Skid base. The pump assembly shall be mounted on a welded skid base of aluminum conforming to ASTM B 308, with rounded ends for skidding. The skid base shall be adequate to maintain alinement of the pump, diesel engine, and all accessories. No component of the pump assembly shall overhang the skid base.

3.14 Lifting bracket. The pump assembly shall have a lifting bracket located at the center of gravity of the pump assembly to enable lifting from above. The lifting bracket shall have an eye of not less than 3 inches inside diameter and shall be at the highest point on the assembly. The lifting bracket shall withstand, without any damage to any part of the pump assembly, 2-1/2 times the resulting force imposed when the pump assembly is suspended by the lifting bracket.

3.15 Tiedown provisions. Permanently affixed tiedown provisions that permit fastening the pump assembly to the floor or deck of a transportation medium shall be provided and shall conform to MIL-STD-209, class 2.

3.16 Electromagnetic interference. The electromagnetic interference emission characteristics of the pump assembly shall conform to MIL-STD-461, class IIIC.

3.17 Lubrication.

3.17.1 Lubricants. The pump assembly shall operate as specified herein when serviced with the following military lubricants:

Engine: MIL-L-2104 or MIL-L-46167.
Grease Lubrication: MIL-G-10924 or MIL-G-23827.

The pump assembly shall be assembled, run-in (broken-in), tested, and delivered with these greases. The bearings shall be cleaned before lubrication, because military greases are not always compatible with other greases. A tag shall be attached in a conspicuous place to indicate which military grease has been used.

3.17.2 Lubrication fittings. Lubrication fittings shall conform to MIL-F-3541, type I, II, or III. Fittings shall be located in a protected position and shall be accessible to a hand-operated grease gun conforming to MIL-G-3859 with a 10-inch flexible extension. Accessibility to fittings shall be provided without the removal or adjustment of accessories or parts. Panels and

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plates equipped with hand-operable quick-disconnect fasteners may be removed to provide accessibility.

3.17.3 Pressure-release device. A pressure-release device shall be provided where the use of pressure lubricating equipment could damage grease seals or other parts.

3.18 Identification marking. The pump assembly shall be identified in accordance with MIL-STD-130. The marking shall be applied to the pump assembly on plates conforming to MIL-P-514, type I, style 1, composition C, of type I, grade A, class 1 material. Plates shall be attached by screws, bolts, or rivets in a location where the plate will be both visible and legible.

3.19 Treatment and painting. The portions of the pump assembly normally painted shall be cleaned, treated, and painted in accordance with MIL-T-704, type A, color forest green.

3.20 Workmanship. The pump assembly shall be free from defects such as misalined components, incomplete welds, cracks, and any other defects that could impair the operating and serviceability characteristics of the pump assembly. All piping, fittings, hoses, and cables shall be supported to prevent damage caused by vibration, chafing, or sharp bends.

3.20.1 Castings and forgings. All parts, components, and assemblies of the pump assembly which include castings and forgings shall be clean of harmful extraneous material such as sand, dirt, sprues, scale, and flux. Rework shall be limited to procedures which do not reduce mechanical properties or effect function.

3.20.2 Metal fabrication. Metal used in fabrication shall be free from kinks and sharp bends. The straightening of material shall be done by methods that will not cause injury to the material. Corners shall be square and true. Flame-cutting, using tips suitable for the thickness of the steel, may be employed instead of shearing and sawing. All bends shall be made with controlled means to insure uniformity of size and shape. Precaution shall be taken to avoid overheating. Heated steel shall be allowed to cool slowly. External surfaces shall be free of burrs, sharp edges, and corners, except when sharp edges or corners are required or where they are not detrimental to safety.

3.20.3 Welding. The surfaces or parts to be welded shall be free from rust, scale, paint, grease, mill scale that can be removed by chipping and wire brushing, and other foreign matter. Welds shall transmit stress without permanent deformation or failure when the parts connected by the welds are subjected to proof and service loading.

3.20.4 Bolted connections. Boltholes shall be accurately formed and shall have the burrs removed. Washers or lockwashers shall be provided where necessary. Matching thread areas securing bolts conforming to SAE J429 or capscrews shall be of sufficient strength to withstand the tensile strength of the bolt. All fasteners shall be correctly torqued and shall have full thread engagement. Bolts shall protrude not more than 2 full threads.

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4. QUALITY ASSURANCE PROVISIONS

4.1 Responsibility for inspection. Unless otherwise specified in the contract, the contractor is responsible for the performance of all inspection requirements as specified herein. Except as otherwise specified in the contract, the contractor may use his own or any other facilities suitable for the performance of the inspection requirements specified herein, unless disapproved by the Government. The Government reserves the right to perform any of the inspections set forth in the specification where such inspections are deemed necessary to assure supplies and services conform to prescribed requirements.

4.1.1 Component and material inspection. The contractor is responsible for insuring that components and materials used are manufactured, examined, and tested in accordance with referenced specifications and standards, as applicable.

4.2 Classification of inspections. The inspection requirements specified herein are classified as follows:

a. First article inspection (see 4.3 and 4.4).
b. Quality conformance inspection (see 4.5).
c. Inspection comparison (see 4.7).
d. Inspection of packaging (see 4.8).

4.3 First article.

4.3.1 Examination. The first article pump assembly shall be examined as specified in 4.6.1. Presence of one or more defects shall be cause for rejection.

4.3.2 Tests. The first article pump assembly shall be subjected to the tests marked "X" in column 1 of table I. Failure of any test shall be cause for rejection.

4.4 Additional inspection. When specified (see 6.2), one or more pump assemblies will be selected at random by the Government from the pump assemblies being produced by production tooling and will be examined as specified in 4.6.1 and subjected to the tests marked "X" in column 1 of table I to determine conformance to the requirements of this specification. The inspection will be performed by the Government at a site selected by the Government. Acceptance of the pump assembly shall not exclude the remaining pump assemblies from the quality conformance inspection and acceptance provisions specified in section 4.

4.4.1 Inspection failure. Failure of the pump assembly to meet any requirements specified herein during and as a result of the examination and tests specified in 4.4 shall be cause for rejection of the pump assembly and shall be cause for refusal by the Government to continue acceptance of production pump assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on the pump assemblies previously accepted and produced under the contract. Any deficiencies found as a result of this inspection will be prima facie evidence that all pump assemblies accepted prior to the completion of this inspection are

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similarly deficient unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.5 Quality conformance inspection.

4.5.1 Sampling. Sampling for examination and tests shall be in accordance with MIL-STD-105.

4.5.2 Examination. Samples selected in accordance with 4.5.1 shall be examined for the major and minor defects specified in 4.6.1. AQL shall be 2.5 percent defective for major defects and 4.0 percent defective for minor defects.

4.5.3 Tests.

4.5.3.1 Individual. Each pump assembly shall be subjected to the tests marked "X" in column 2 of table I. Failure of any test shall be cause for rejection.

4.5.3.2 Samples. Samples selected in accordance with 4.5.1 shall be subjected to test marked "X" in column 3 of table I. AQL shall be 2.5 percent defective.

4.6 Inspection procedure.

4.6.1 Examination. The pump assembly shall be examined as specified herein for the following defects:

Major

101. Materials not as specified.
102. Materials are not corrosion resistant or treated to be made corrosion resistant for the applicable storage and operating environments.
103. Dissimilar metals as defined in MIL-STD-889 are not effectively insulated from each other.
104. Contractor does not have documentation available for identification of material, material finishes or treatments.
105. Threads not as specified.
106. Pump case not as specified.
107. Pump impeller not as specified.
108. Impeller balancing not as specified.
109. Wearing rings not as specified.
110. Pump suction port not as specified.
111. Pump discharge port not as specified.
112. Check valve assembly not as specified.
113. Cranking system not as specified.
114. Charging system not as specified.
115. Performance identification plate missing or not as specified.
116. Components missing or not as specified.
117. Instruments missing or not as specified.
118. Manual engine controls missing or not as specified.
119. Skid base not as specified.
120. Lubricants not as specified.

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121. Lubrication fittings not as specified.
122. Pressure-release device not as specified.
123. Workmanship not as specified.
124. Castings and forgings not as specified.
125. Metal fabrication not as specified.
126. Welding not as specified.
127. Bolted connections not as specified.

Minor

201. Identification marking missing, incorrect, or illegible.
202. Treatment and painting not as specified.

4.6.2 Tests.

4.6.2.1 Test conditions. Unless otherwise specified in a test, tests shall be performed in accordance with applicable provisions of the test code for centrifugal pumps of the standards of the Hydraulic Institute. All performance data shall be corrected to standard sea level conditions of barometric pressure of 29.92 inches of mercury. Water at a temperature of between 60° F and 80° F may be used as the test fluid.

4.6.2.2 Test schedule.

TABLE I. Test schedule.

First article	Quality Conformance: Individual	Quality Conformance: Sample	Tests	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X			High temperature.	4.6.2.3	3.5
X			Low temperature.	4.6.2.4	3.5
X			Safety.	4.6.2.5	3.6
X			Human factors.	4.6.2.6	3.7
X			Noise level measurement.	4.6.2.6.1	3.7.1
X			Reliability.	4.6.2.7	3.8
X			Maintenance support.	4.6.2.8.1	3.9.1
X			Maintenance ratio.	4.6.2.8.2	3.9.2
			Performance:		
X			Pump	4.6.2.9.1	3.10
X		X	Operational	4.6.2.9.2	3.10 & 3.12.2

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TABLE I. Test schedule. (continued)

First article	Quality Conformance: Individual	Quality Conformance: Sample	Tests	Test paragraph	Requirement paragraph
1	2	3	4	5	6
X	X		Hydrostatic pressure.	4.6.2.10	3.11.1
X	X		Impeller balance.	4.6.2.11	3.11.3.1
X		X	Priming.	4.6.2.12	3.11.9
X	X		Engine.	4.6.2.13	3.12
X	X		Fuel tank.	4.6.2.14	3.12.3
X			Lifting bracket.	4.6.2.15	3.14
X			Tiedown provisions.	4.6.2.16	3.15
X			Electromagnetic.	4.6.2.17	3.16

4.6.2.3 High temperature. The pump assembly shall be tested as specified in MIL-STD-810, procedure II, method 501.1. The maximum ambient storage temperature shall be 160° F and the maximum ambient operating temperature shall be 125° F. The operating period of procedure II shall be 2 hours. The pumping assembly shall be operated at rated capacity. Noncomformance to 3.5 shall constitute failure of this test.

4.6.2.4 Low temperature. The pump assembly shall be tested as specified in MIL-STD-810, procedure I, method 502.1. The minimum ambient storage temperature shall be -50° F. The minimum ambient operating temperature shall be +32° F. The operating period of procedure I shall be 2 hours. The pump assembly shall be operated at rated capacity. Nonconformance to 3.5 shall constitute failure of this test.

4.6.2.5 Safety. The pump assembly shall be evaluated for safety requirements throughout testing as specified in 4.6.2. Nonconformance to 3.6 shall constitute failure of this test.

4.6.2.6 Human factors engineering. The pump assembly shall be evaluated for human factors engineering requirements throughout testing as specified in 4.6.2. Nonconformance to 3.7 shall constitute failure of this test.

4.6.2.6.1 Noise level test. Noise levels shall be measured in accordance with MIL-STD-1474, requirements for steady state noise, and reported in the format indicated by MIL-STD-1474, figure 7. As a minimum: Noise levels shall be measured when equipment is operating under full load, in accordance with

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MIL-STD-1474, requirements for noise contours. Contours shall be taken at not fewer than 12 equal (horizontal) arc increments, one increment shall include data from the noisiest position. Additionally, the noise level at the operator's typical operating position shall be provided as dB(A) level. Should the noise level exceed 85 dB(A) at any noise contour point or the operator's position, failure to comply with 3.7.1 or MIL-STD-1474 provisions shall constitute failure of this test.

4.6.2.7 Reliability. Using MTBF specified in 3.8, the pump assembly shall be tested as specified in 4.6.2.9.2 with "accept" and "reject" criteria in accordance with MIL-STD-781, test plan IV. Continue testing until either an "accept" or "reject" decision is reached. A failure is defined as any malfunction which cannot be corrected within 30 minutes (all waiting periods are excluded by adjustment, repair, or replacement action using controls and on-equipment tools or parts available to the using element, and is a malfunction which causes or may cause one or more of the following:

a. Failure to commence operation, cessation of operation, or degradation of performance below specified levels.
b. Damage to the pump assembly by continued operation.
c. Safety hazards to personnel.

Nonconformance to 3.8 shall constitute failure of this test.

4.6.2.8 Maintainability.

4.6.2.8.1 Maintenance support evaluation. Maintenance operations required during first article testing shall be accomplished to determine conformance to 3.9.1. Failure of the support elements to maintain the pump assembly as specified shall constitute failure of this test. Errors or inadequacies in the manual shall not be considered in assessing the maintenance ratio or the MTBF.

4.6.2.8.2 Maintenance ratio. The maintenance ratio shall be computed during first article testing. Nonconformance to 3.9.2 shall constitute failure of this test.

4.6.2.9 Performance.

4.6.2.9.1 Pump. Prior to assembling the pump to the engine, operate the pump using a dynamometer or calibrated electric motor at the designed operating speed and at a static suction life equivalent to 10 feet of water at standard conditions. Test the pump for the requirements specified in 3.10 in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head capacity shall be taken at a minimum of six points and shall include the rating points (see 3.10), point of maximum operating speed, wide-open discharge, and shut-off. Delivery of less than the capacities specified in 3.10 shall constitute failure of this test.

4.6.2.9.2 Pump assembly (operational). Assemble the pump to the engine and operate the pump at the designed operating speed and at a static suction lift equivalent to 10 feet of water at standard conditions. Test the pump for

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practicable. Provide the pump and pump assembly product tests all other

requirements specified in 3.10 in accordance with the Rating Standards and Test Code Standards for centrifugal pumps of the Hydraulic Institute. Data relative to head and capacity of the pump and speed (rpm) of the engine shall be measured and recorded at not less than six points on the pump performance curve. The measured points shall include the rating points (see 3.10), point of maximum operating speed, wide-open discharge, and shut-off. The pump assembly shall run for not less than 1 minute at each measured point. Delivery of less than the capacities specified in 3.10 or failure of the engine to meet any of the operational requirements or conditions specified in 3.12.2 shall constitute failure of this test.

4.6.2.10 Pump hydrostatic pressure. With check valve assembly installed, subject the pump to an internal hydrostatic pressure of not less than 260 psi for a period of not less than 3 minutes. Leakage or permanent deformation of any component of the pump or check valve assembly shall constitute failure of this test (3.11.1).

4.6.2.11 Pump impeller. Determine the out-of-balance condition of the impeller while rotating at the continuous operating speed of the pump assembly. An out-of-balance condition of more than 0.25 oz-in shall constitute failure of this test.

4.6.2.12 Priming. The pump assembly shall prime and operate under the conditions specified herein. Nonconformance to the requirements of 3.11.9 shall constitute failure of this test.

4.6.2.13 Engine performance. Prior to installation of the engine to the pump and examination tests of the first article, the engine shall be tested in accordance with MIL-STD-1400, test method 2000. Nonconformance to the requirements specified herein shall constitute failure of this test.

4.6.2.14 Fuel tank. With all attachments installed, with all openings plugged or capped, apply an internal air pressure of 5 pounds per square inch gage (psig) to the fuel tank and immerse in water so that the tank top is not less than 12 inches under the water level. Any evidence of leakage from the tank shall constitute failure of this test.

4.6.2.15 Lifting bracket. Place the pump assembly on a flat surface and lift it with the lifting bracket. Raise the pump assembly until the skid is completely cleared of the ground. With the pump assembly in this position, add weights to the pump assembly until the force on the lifting bracket is 2-1/2 times the initial force. Hold the pump assembly under this load condition for not less than 90 seconds. Lower the pump assembly onto the surface and examine for damage. Nonconformance to 3.14 shall constitute failure of this test.

4.6.2.16 Tiedown provisions.

4.6.2.16.1 Test procedure. With the pump assembly anchored by means other than the tiedown being tested, apply the static loads specified in MIL-STD-209, class 2 to each provision. Measure and record the dimensions of the tiedowns.

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4.6.2.16.2 Failure criteria. Weld failure, permanent deformation of the pump assembly or its tiedown provisions, breakage of the provisions, or nonconformance to MIL-STD-209 shall constitute failure of this test.

4.6.2.17 Electromagnetic interference. The first article shall be tested to determine conformance to 3.16. The contractor shall furnish the contracting officer a report of tests required by MIL-STD-461. Disapproval of the report shall constitute failure of this test.

4.7 Inspection comparison. The Government may select pump assemblies at any time during the contract production period and subject these pump assemblies to the examination specified in 4.6.1 and the test specified in 4.6.2 to determine conformance to the requirements of this specification. The inspection will be performed by the Government, at a site selected by the Government, on pump assemblies selected at random from those which have been accepted by the Government and will not include the previously inspected first article assemblies. In addition to any test specified as part of the inspection comparison, the Government reserves the right to conduct any and all other tests contained in this specification as part of the inspection comparison, and failure of such additional tests shall have the same effect as failure of those tests specified as inspection comparison.

4.7.1 Inspection failure. Failure of an inspection comparison pump assembly to meet any requirement specified herein during and as a result of the examination and tests specified in 4.7 shall be cause for rejection of the inspection comparison pump assembly and shall be cause for refusal by the Government to continue acceptance of production pump assemblies until evidence has been provided by the contractor that corrective action has been taken to eliminate the deficiencies. Correction of such deficiencies shall be accomplished by the contractor at no cost to the Government on pump assemblies previously accepted and produced under the contract. Any deficiencies found as a result of the inspection comparison will be considered prima facie evidence that all pump assemblies accepted prior to the completion of inspection are similarly deficient, unless evidence to the contrary is furnished by the contractor and such evidence is acceptable to the contracting officer.

4.8 Inspection of packaging.

4.8.1 First article inspection of packaging. The first article packaging shall be examined and tested in accordance with the first article pack (preproduction pack) inspection provisions of MIL-P-10603.

4.8.2 Quality conformance inspection. Packaging of pump assemblies shall be examined in accordance with the quality conformance inspection provisions of MIL-P-10603.

5. PACKAGING

5.1 Packaging. Packaging of the pumps shall be level A, level B, or commercial as specified (see 6.2). Preservation materials and their application shall not adversely affect the potability of water to be pumped through this assembly.

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5.1.1 Levels A and B. Levels A and B packaging, which includes preservation, packing and marking, shall be in accordance with the applicable provisions for levels A and B of MIL-P-10603.

5.1.2 Commercial. Commercial packaging, which includes preservation, packing and marking, shall be in accordance with ASTM D 3951. Additionally, each shipping container shall be marked with the gross weight and cube.

5.2 Depreservation guide. Two depreservation guides shall be prepared. One guide shall be placed in a waterproof envelope marked "Depreservation Guide" and shall be secured to the pump assembly in a conspicuous location. The second guide shall be preserved with the technical publications and the bag shall be marked to indicate that the depreservation guide is inside. Unless otherwise specified (see 6.2), DA Form 2258, "Depreservation Guide for Vehicles and Equipment", shall be used.

6. NOTES

6.1 Intended use. The pump assembly is intended for general purpose water-pumping service.

6.2 Ordering data. Acquisition documents should specify the following:

a. Title, number, and date of this specification.
b. Time frame and number of assemblies required for submission of first article (see 3.2).
c. When the Government will conduct any or all of the first article examination and tests. When the Government will conduct some but not all of the first article examination and tests, the contracting officer should specify which examination and tests will be conducted by the Government and which examination and tests will be conducted by the contractor (see 3.2).
d. When additional inspection is required and number of pump assemblies to be furnished (see 4.4).
e. Degree of packaging required (see 5.1).
f. When other than DA Form 2258 shall be used (see 5.2).

6.3 First article. When a first article inspection is required, the items should be a preproduction model. The first article should consist of one or more units. The contracting officer should include specific instructions in acquisition documents regarding arrangements for examinations, tests and approval of the first article test results and disposition of the documents first article.

6.4 Data requirements. The contracting officer should include requirements for such data as technical publications, instructional materials, illustrated parts lists, and contractor's maintenance and operation manual to be furnished with each pump assembly.

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6.5 Provisioning. The contracting officer should include provisioning requirements for repair parts and maintenance tools, as necessary (including any special tools), and instructions regarding shipment of pump assemblies.

6.6 Depreservation guides. The contracting officer should arrange to furnish the necessary copies of DA Form 2258, when requested by the contractor (see 5.2).

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STANDARDIZATION DOCUMENT IMPROVEMENT PROPOSAL

(See Instructions – Reverse Side)

1. DOCUMENT NUMBER	2. DOCUMENT TITLE
MIL-P-52469B(ME)	Pump, Centrifugal, Water, DED, 2-½-In, Skid-Mtd, 200 GPM at 300 Ft Total Head Auxiliary Pri

3a. NAME OF SUBMITTING ORGANIZATION	4. TYPE OF ORGANIZATION *(Mark one)*
	☐ VENDOR ☐ USER ☐ MANUFACTURER ☐ OTHER *(Specify)*: _____
b. ADDRESS *(Street, City, State, ZIP Code)*	

5. PROBLEM AREAS

a. Paragraph Number and Wording:

b. Recommended Wording:

c. Reason/Rationale for Recommendation:

6. REMARKS

7a. NAME OF SUBMITTER *(Last, First, MI)* – Optional	b. WORK TELEPHONE NUMBER *(Include Area Code)* – Optional
c. MAILING ADDRESS *(Street, City, State, ZIP Code)* – Optional	8. DATE OF SUBMISSION *(YYMMDD)*

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